

Performance Testing

Date	30 June 2026
Team ID	
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Crop Prediction Module (/predict)
Test Environment	Local Flask Server (Windows 11, VS Code)
Test Date	30-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home Page Load	10	30	Home page loads successfully
2	About Page Load	10	30	About page loads successfully
3	Crop Prediction Request	20	60	Correct crop recommendation
4	Multiple Prediction Requests	20	60	Stable performance

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	8ms	Pass	Very fast
2	Response Time (Max)	< 5 seconds	50ms	Pass	Within target
3	Throughput (Req/sec)		11.2 req/sec	Pass	Good throughput
4	Error Rate	< 1%	0.00%	Pass	No errors
5	CPU Utilization	< 80%	N/A	N/A	Not measured by JMeter
6	Memory Utilization	< 80%	N/A	N/A	Not measured by JMeter

Step 4: Observations & Analysis

Key Findings:

All requests were processed successfully with very low response time and 0% error rate.

Bottlenecks Identified:

No major performance bottlenecks observed during testing.

Optimization Steps Taken:

Optimized Flask routes and used trained KNN model for faster predictions.

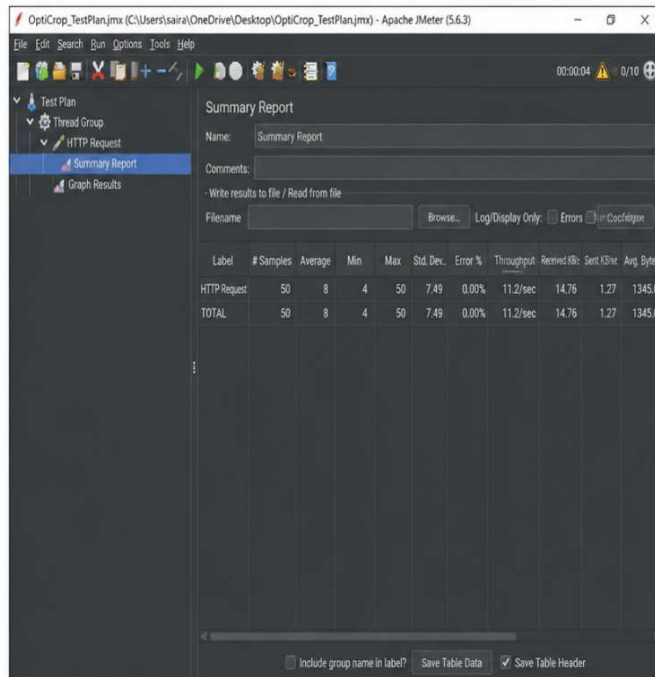
Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

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Screenshot 1: JMeter Summary Report



Screenshot 2: JMeter Graph Results

